**Project Design Phase**
**Problem – Solution Fit Template**

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| Date | 12 October 2026 |
| Team ID | SWUID2026403463 |
| Project Name | HotelPro - Online Hotel Room Booking & Reservation Management System |
| Maximum Marks | 2 Marks |

## Problem – Solution Fit Overview:

The Problem–Solution Fit ensures that the HotelPro platform addresses real gaps in how people discover and book hotel rooms and stays, and aligns with the needs of guests, hotel managers and administrators. This validation is critical before the system is built and scaled.

## Purpose:

- Bridge the gap between travellers who want a quick, reliable way to find and book rooms, and hotels that want to fill them.
- Offer one unified platform covering the whole stay journey, from discovery and room selection to payment, booking voucher and cancellation, without juggling several apps, calls or front-desk visits.
- Prevent double bookings and payment confusion through room-inventory locking, secure checkout and clear booking statuses.
- Give hotel managers and admins tools to manage rooms, rates, offers and revenue with live occupancy and analytics.
- Improve booking success and guest satisfaction through destination-based search, room galleries, reviews, recommendations and timely notifications.

## Problem Statement:

Many travellers, hotel operators and front-desk teams face challenges such as:

- Difficulty comparing hotels, room types, rates and availability in one place for a destination
- Two guests booking the same room for the same dates, leading to failed or conflicting reservations
- Phone and front-desk bookings that are slow, error-prone and limited to working hours
- Slow, unclear cancellations and refunds with no visibility of status
- Limited tools for hotel managers to track occupancy, bookings and revenue
- Poor personalization, with no easy way to save favorites, read reviews or get reminders

## Solution:

**HotelPro**, a full-stack hotel booking platform, offers:

- Destination-based discovery of hotels and rooms with search, auto-complete, filters and sorting
- Real-time room availability calendar with a 10-minute room hold that prevents double booking
- End-to-end booking flow with add-ons (breakfast, airport pickup, spa), coupons, order summary and mock payment gateway
- QR-code booking vouchers that front-desk staff can verify, plus a My Bookings dashboard
- Policy-based cancellation (up to 24 hours before check-in) with automatic refunds and status tracking
- Room-type management with rate plans, seasonal pricing, inventory control and manager sales dashboards
- Reviews, ratings, favorites, recommendations and email/in-app notifications
- Role-based admin, hotel manager and guest portals with analytics, audit logs and secure JWT-based access